



D(6th Sm.)-Computer Science-H/DSE-A-4/CBCS

2025

COMPUTER SCIENCE — HONOURS

Paper : DSE-A-4

(Multimedia and Its Applications)

Full Marks : 50

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer **question no. 1** and **any four** from the rest.

1. Answer **any five** questions :

2×5

- (a) Name four major components of multimedia.
- (b) Briefly discuss the role of entropy in multimedia.
- (c) State two differences between MIDI and Digital Audio.
- (d) What is the purpose of a colour palette in digital imaging?
- (e) Define Keyframe in Animation.
- (f) Differentiate between lossless and lossy audio compression.
- (g) Name two commonly used video file format and their use cases.
- (h) Define the term frame rate in video processing.

2. (a) Explain how file size of a digital audio impacts the quality of the audio.

(b) A frame buffer is used to store images at a resolution of 1920×1080 pixels, with each pixel requiring 24 bits of colour information. Calculate the total amount of memory (in bytes) required to store a single frame.

(c) Calculate the required bandwidth in bytes per second to display a 1920×1080 resolution video at 60 Hz with 24 bits per pixel. 4+3+3

3. (a) Discuss the process of digitizing an audio signal.

(b) What are the key requirements for real-time multimedia communication over networks? 6+4

4. (a) Explain the role of a multimedia authoring tool. What are some commonly used authoring tools?

(b) Discuss the role of graphics processing units (GPUs) in multimedia. How do they differ from CPUs in this context? (3+2)+(3+2)

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5. (a) State two advantages and two disadvantages of MIDI files over Digital audio.
(b) Describe 2D, 2.5D and 3D animation. 4+6
6. (a) What are the differences between vector-drawn objects (vector image) and bitmap (bitmap image)?
(b) Explain why vector images achieve higher quality while scaling than bitmap images. 5+5
7. (a) Differentiate between analog and digital signal.
(b) Differentiate between key-frame and path animation.
(c) Discuss the steps to create a key-frame animation with appropriate diagrams. 2+4+4
8. (a) Describe the impact of multimedia on advertising and marketing strategies.
(b) Discuss the ethical considerations in using multimedia content. 5+5
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